MR compatible microscope assembly instructions

General notes before starting assembly

- We tried to **avoid using glue** as much as possible. Some glues have an effect on MR signal, and decrease the ease of customization/replace parts that are damaged. We recommend using **NOA61 glue** for the reasonable resilience and optical properties (https://www.thorlabs.com/thorproduct.cfm?partnumber=NOA61). In general, please use low amounts of glue to prevent glue on optical components, and always cure from all directions!
- **Lenses should have a flush fit**. However, when a lens is to lose, try to fix it in place with minimal optical glue.
- Please **ensemble the microscope in a clean room** to avoid small debris and dust from entering the optical system.
- Our MR compatible microscope is **optimized for GCaMP and RCaMP imaging**. If you require additional/different fluorophores, consider replacing the filters and **leaving a message on the Git message board** with the results!
- The objective in an **immersion objective optimized for D2O imaging**. When constructing the microscope, please consider using D2O to immerse your target when optimizing the focus. Alternatively, regular water would work.

Instructions

Please use the following instructions with the illustrations on the pages bellow!

Icons:



Additional information video available in instructions.



Assembly requires glue, please minimize glue usage when unnecessary.

1. Slide ***lens_15271*** into the ***microscope_objective***, and place the wole part into ***construction_objective***. The lens should rest on the bottom part of the ***construction_objective***. Consider placing one protective lens cleaning tissue sheets bellow the ***lens_15271***.
2. Glue the lens into place using optical glue. Start with two small drops in the front and back, and cure with a UV-light source. Once the glue is cured, gently apply glue around the objective to ensure no liquid can enter during immersion imaging. Cure the newly applied glue, and leave for at least 24 hours to completely settle.
3. Place ***Lens_36629*** (2x) and ***microscope_spacer*** (1x) into ***microscope_objective*** from the top.
4. Again, glue the lens into place using optical glue. Start with two small drops in the front and back, and cure with a UV-light source. Once the glue is cured, gently apply glue around the objective to ensure no dust can enter. Cure the newly applied glue, and leave for at least 24 hours to completely settle.

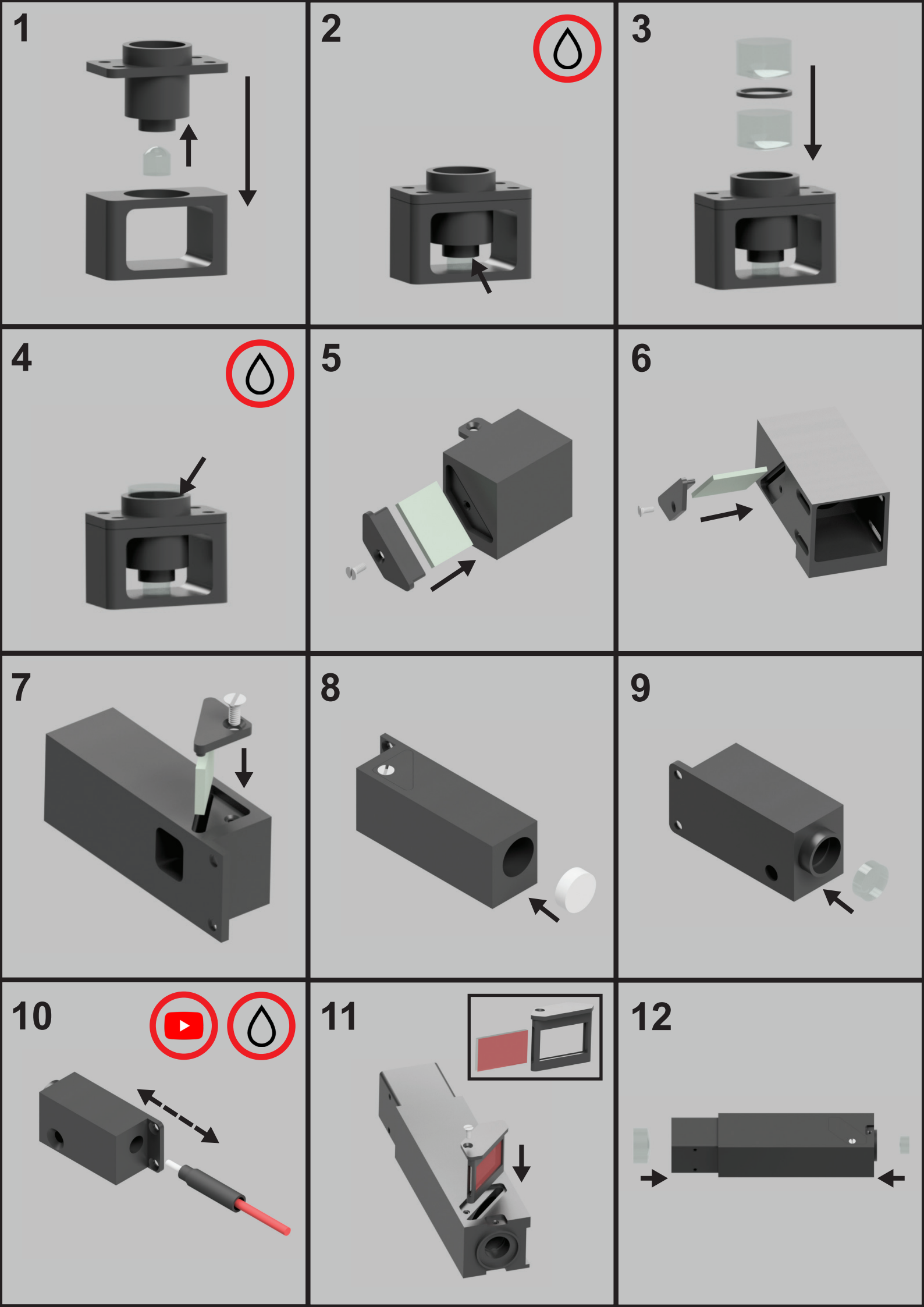
Note: step 9 and 10 also require waiting for glue to completely cure. Consider doing these steps at the same time as step 1-4.

5. Slide the ***mirror_mirror8x11*** into the ***microscope_objectivecoupler***. Make sure the mirror is facing the inside of the ***microscope_objectivecoupler***. Next, seal the mirror with the ***microscope_mirrorlockfront*** and one 2.5 mm M1.2 x 0.25 PEEK screw.
6. Slide the ***mirror_mirror16x11*** into the ***microscope_cameracoupler***. Make sure the mirror is facing the inside of the ***microscope_cameracoupler***. Next, seal the mirror with the ***microscope_mirrorlockback*** and one 2.5 mm M1.2 x 0.25 PEEK screw.
7. Slide the ***mirror_mirror5x8*** into the ***adaptor_lightsourceangle***. Make sure the mirror is facing the inside of the ***adaptor_lightsourceangle.*** Next, seal the mirror with the ***microscope_mirrorlocktop*** and one 2.5 mm M1.2 x 0.25 PEEK screw.
8. Slide the ***filter_diffuser*** into the ***adaptor_lightsourceangle***. If absolutely necessary, gently fix in place with optical glue and let cure for 24 hours to avoid vapor residue on other optical parts.
9. Slide the ***lens_63594*** into the ***adaptor_lightsource***.
10. Couple the end of the ***excitation fiber*** into an appropriate light source (any light source for GCaMP (~470 nm) or RCaMP (~560 nm) imaging works. Place the ***adaptor_lightsource*** approximately 5 cm in front of a white sheet of paper/white wall. Carefully slide in the ***excitation fiber***, and fix the fiber with optical glue at the point where the output beam is collimated. To optimize the collimation, please follow this online tutorial: https://www.youtube.com/watch?v=z_n7GKdTt0Q.
11. Slide the ***filter_dichroic*** into the ***microscope_filtercube***, and fix the ***microscope_filtercube*** into ***microscope_opticaltube*** with 2.5 mm M1.2 x 0.25 PEEK screw.
12. Slide ***lens_47662*** (left, large diameter lens), and ***lens_47690*** (right, small diameter lens) into ***microscope_opticaltube***.
13. Slide the ***filter_emmiter*** into the ***adaptor_camera***. In addition, place the ***adaptor_cameralock*** on the camera.
14. Try to align the ***adaptor_cameralock*** with the edge of the camera. Next, screw the **adaptor_camera** into the camera, but do not completely lock at this stage.

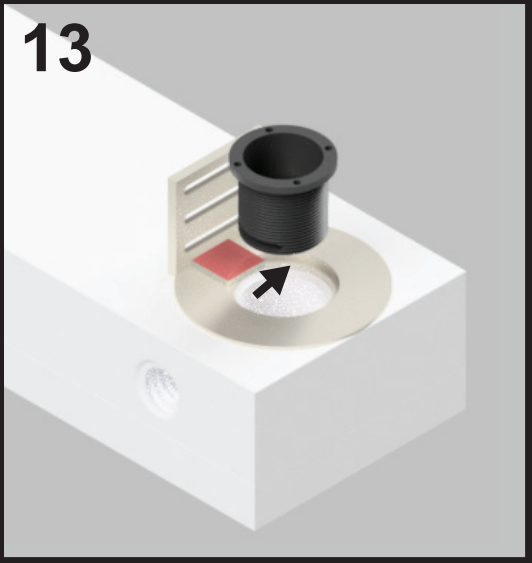
Note: The adaptor camera and adaptor cameralock can be redesigned to fit other MR compatible cameras. However, please keep the distance between the mirror (***mirror_mirror16x11***) in the ***microscope_cameracoupler*** and the camera chip fixed.

15. Attach the ***microscope_objectivecoupler*** to the front of the ***microscope_opticaltube.***
16. Attach the ***microscope_objectivecoupler*** with 2.5 mm M1.2 x 0.25 PEEK screws.
17. Attach the ***microscope_objective*** with 2.5 mm M1.2 x 0.25 PEEK screws to the bottom of the ***microscope_objectivecoupler***
18. Slide the ***adaptor_lightsource*** into the ***adaptor_lightsourceangle***.
19. Attach the combined ***adaptor_lightsource*** and ***adaptor_lightsourceangle*** to the ***microscope_opticaltube*** with four 2.5 mm M1.2 x 0.25 PEEK screw and one 10 mm M2 x 0.4 PEEK screw. Start with the 10 mm M2 x 0.4 PEEK screw to hold the parts in place.
20. Slide the ***microscope_cameracoupler*** into the ***microscope_opticaltube***.
21. Attach the ***microscope_cameracoupler*** with two 2.5 mm M1.2 x 0.25 PEEK screws. The distance between these two parts is later used to focus the image, so do not fasten the screws fully during this step.

22. Fix the ***microscope*** to the MR compatible camera:
    a) Slide the build ***microscope*** into the ***adaptor_camera***.
    b) Adjust the ***adaptor_cameralock*** so the ***microscope*** is positioned straight along the longitudinal axis of the camera.
    c) Remove the ***microscope***, and fully fasten the ***adaptor_camera***.
    d) Slide the ***microscope*** back into the ***adaptor_camera***.
23. Attach the ***microscope*** into ***adaptor_cameralock*** with M1.2 x 0.25 PEEK screws
24. Slide the ***microscope_support_lock*** part into the ***microscope_support***. Next, attach the ***support*** to the ***microscope_opticaltube*** with 2.5 mm M1.2 x 0.25 PEEK screws. Fully fasten the back screws, but leave the front screws unfastened. In addition, use the screw facing the camera to make the top of the ***microscope_support*** flush with the **microscope**. The support part is a weight-bearing beam that increases the durability of the ***microscope***, especially when used with MR compatible cameras that run hot.
25. Finally, focus the **microscope** on an object (for example a resolution target test: https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=4338). For this step it is recommended to use D2O to get the optimal focus, however H2O would work as well. Adjust the distance between the ***microscope_opticaltube*** and the ***microscope_cameracoupler*** till the image is sharp. Once the image is sharp, fasten the remaining 2.5 mm M1.2 x 0.25 PEEK screws.
26. Try imaging different samples!


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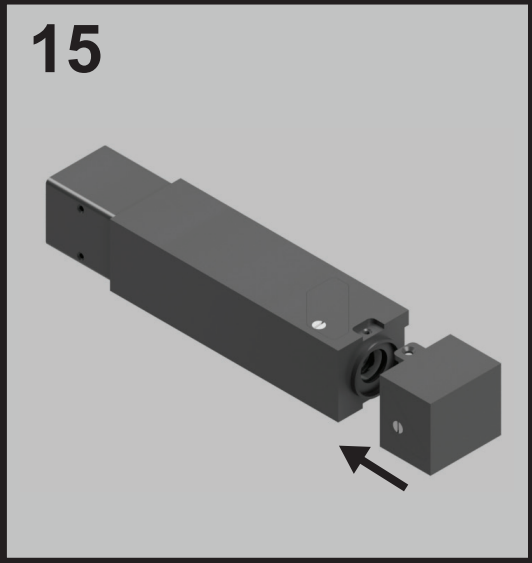
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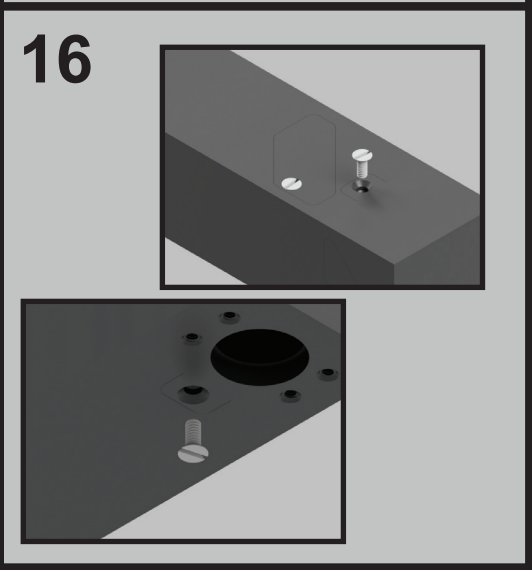
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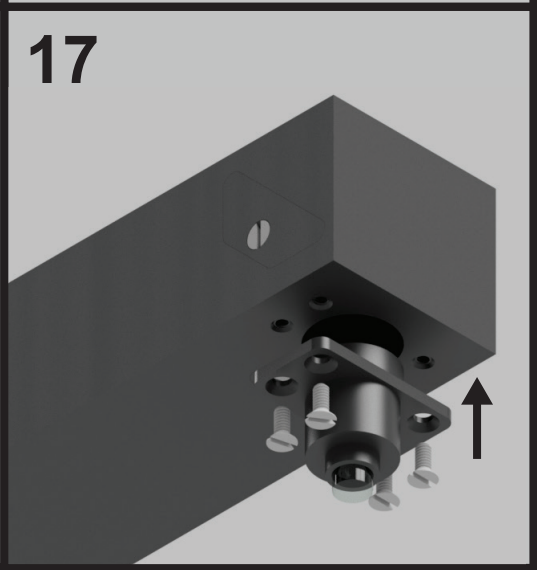
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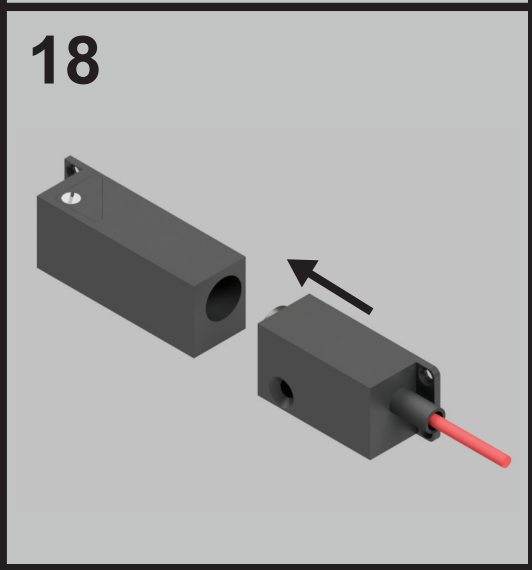
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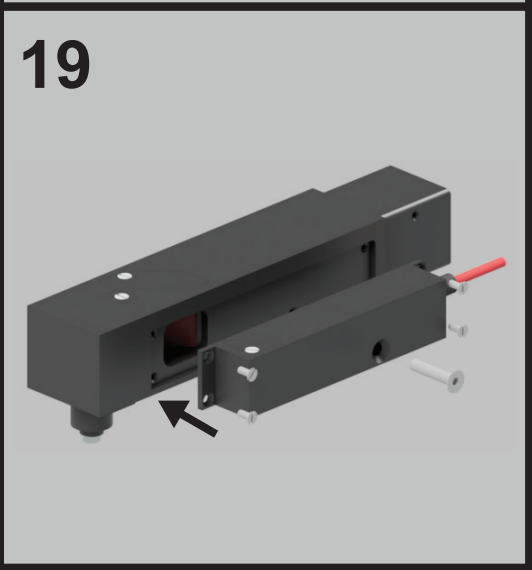
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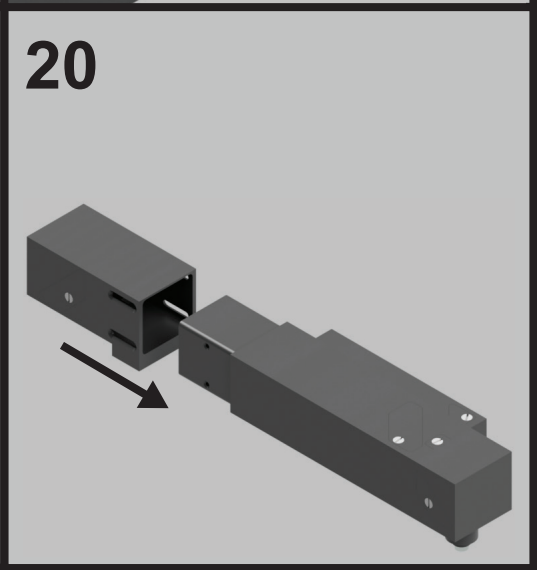
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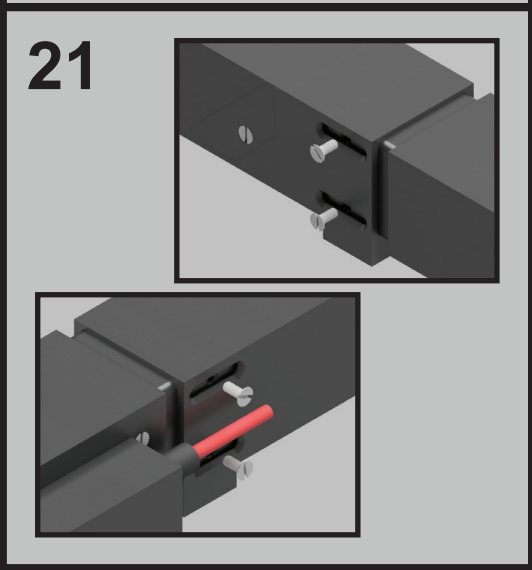
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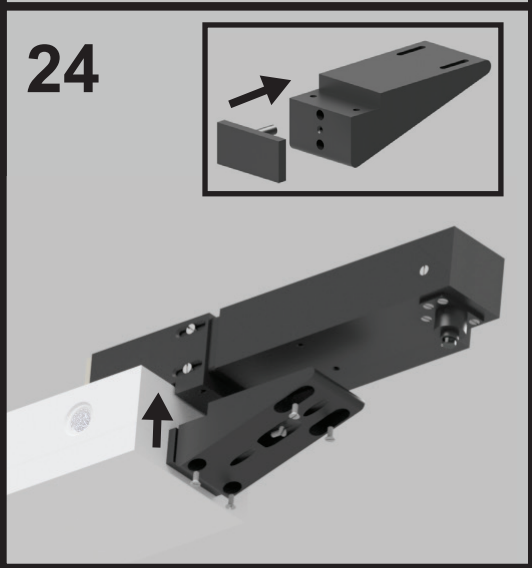
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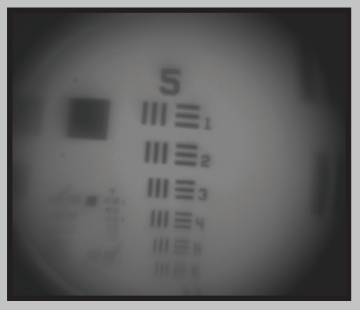


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